WASP-49b

R-1000

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-49_241026` · created 2026-07-30T01:15:48+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460609.9668 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.014
Transit depth (Rp/Rs)^2	1.95 +/- 0.4 [%]
Orbital Inclination (inc)	84.14 +/- 0.62
Airmass coefficient 1 (a1)	0.9457 +/- 0.0072
Airmass coefficient 2 (a2)	0.035 +/- 0.024
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	#1 - [61, 52]
Optimal Aperture	3.59
Optimal Annulus	9.03
Transit Duration (day)	0.0764 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.14 ± 0.014 vs 0.116 (deviation 1.29σ)
Duration fit vs published	0.0764 d vs 0.09 d (deviation 1.6σ)
Mid-time O–C	-12.7 min
Depth SNR	7.5
Residual scatter	0.76 %

Light curve

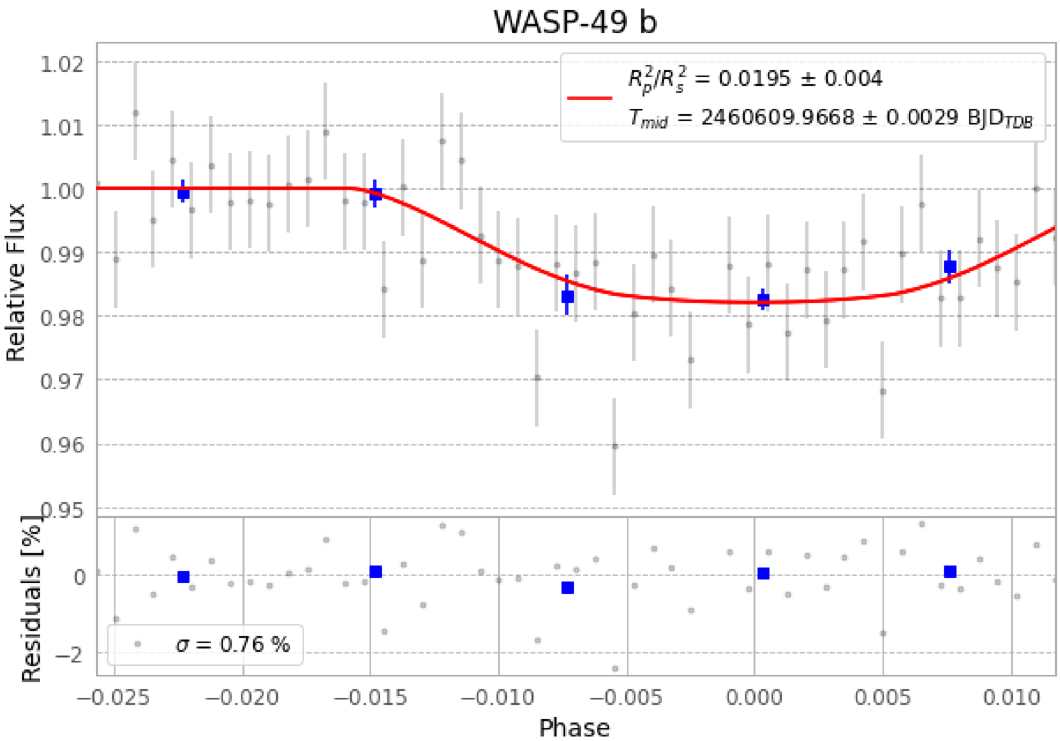

Figure 1 — FinalLightCurve_WASP-49 b_2024-10-26.png (SHA-256 ac29d40098f9a663…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [517, 198], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 57e761b5f701e79b…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

51 FITS frames (33 MB), `WASP-49241026092415.FITS` → `WASP-49241026115415.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-241026.log (f7b2b77388d0…), FinalLightCurve_WASP-49 b_2024-10-26.png (ac29d40098f9…), FinalLightCurve_WASP-49 b_2024-10-26.csv (607fde2ee0c4…)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-30 01:15Z.